

ML-KEM.KeyGen()

Generates an encapsulation key and a corresponding decapsulation key.

Output: encapsulation key $ek \in \mathbb{B}^{384k+32}$.

Output: decapsulation key $dk \in \mathbb{B}^{768k+96}$.

1: $d \xleftarrow{\$} \mathbb{B}^{32}$

▷ d is 32 random bytes

2: $z \xleftarrow{\$} \mathbb{B}^{32}$

▷ z is 32 random bytes

3: **if** $d == \text{NULL}$ **or** $z == \text{NULL}$ **then**

4: **return** $\perp$

▷ return an error indication if random bit generation failed

5: **end if**

6: $(ek, dk) \leftarrow \text{ML-KEM.KeyGen_internal}(d, z)$

▷ run internal key generation algorithm

7: **return** (ek, dk)